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Deep Learning Enhanced SLAM II:
Back-End Optimization and Semantic SLAM

Graduate Course INTR-6000P

Week 11 - Lecture 22

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Semantic SLAM

- Combine geometry + semantics
- Tasks: object detection, segmentation, dynamic object handling

Benefits of Semantic SLAM

- Improved tracking and mapping
- Scene understanding for navigation
- Robustness to dynamic objects

- **Through SLAM to improve semantic recognition.**
- **Through semantic recognition to improve SLAM.**

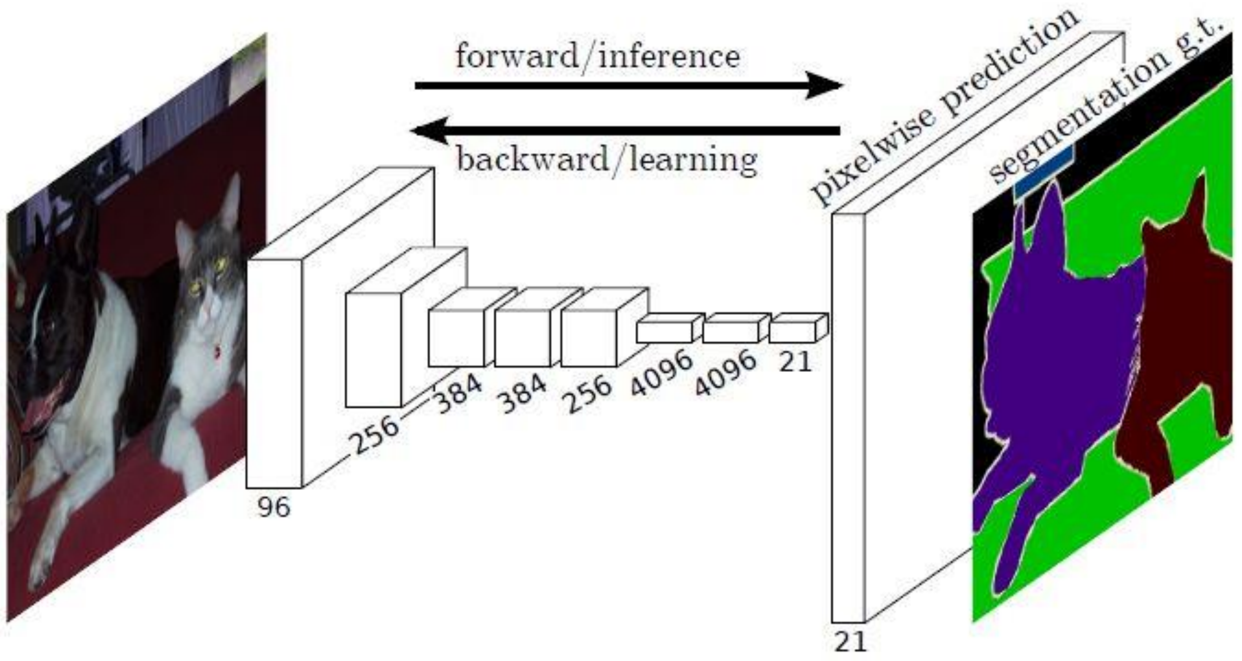
Semantic SLAM

Semantic Segmentation

Fully Convolutional Network (FCN)

Input the RGB image into a convolutional neural network to extract feature maps through multiple convolution and pooling layers.

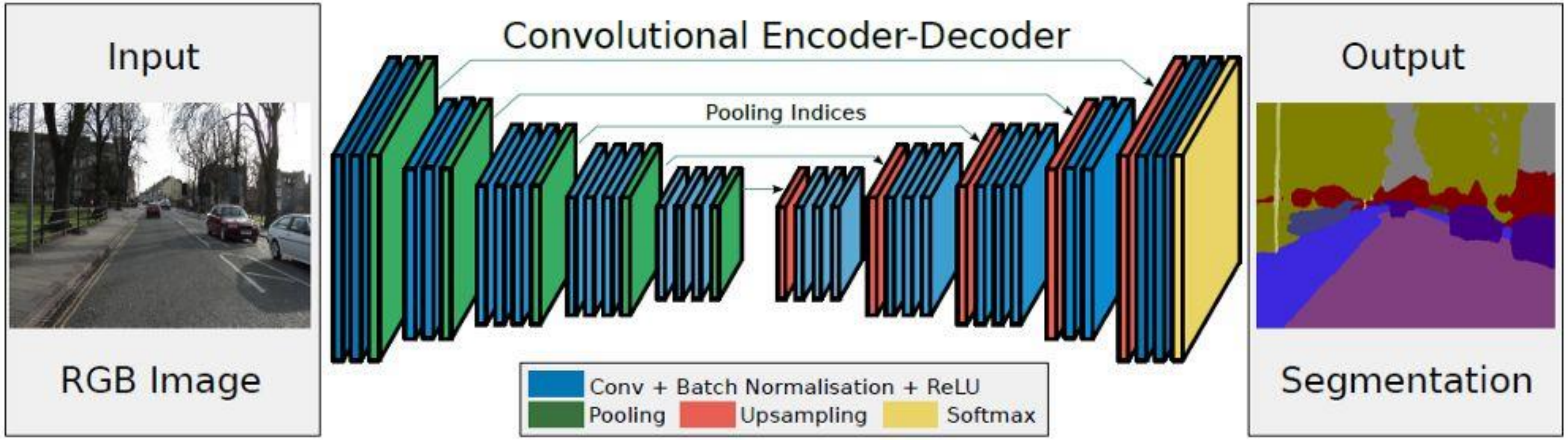
- Use transposed convolution (deconvolution) layers to upsample the final convolutional feature map so that it matches the original image resolution, preserving spatial location information.
- Perform pixel-wise classification on the upsampled feature map, computing the softmax loss for each pixel.



Semantic SLAM

Semantic Segmentation

- **SegNet**
- Addresses the issues of fixed receptive fields and the loss or smoothing of object details in semantic segmentation found in FCN.
- The encoder primarily consists of the first 13 convolutional layers and 5 pooling layers of the VGG16 network, while the decoder similarly comprises 13 convolutional layers and 5 upsampling layers. The high-dimensional features output by the final decoder are fed into a trainable softmax classifier to classify each individual pixel.
- The SegNet network utilizes pooling indices to preserve image contour information and reduce the number of parameters.

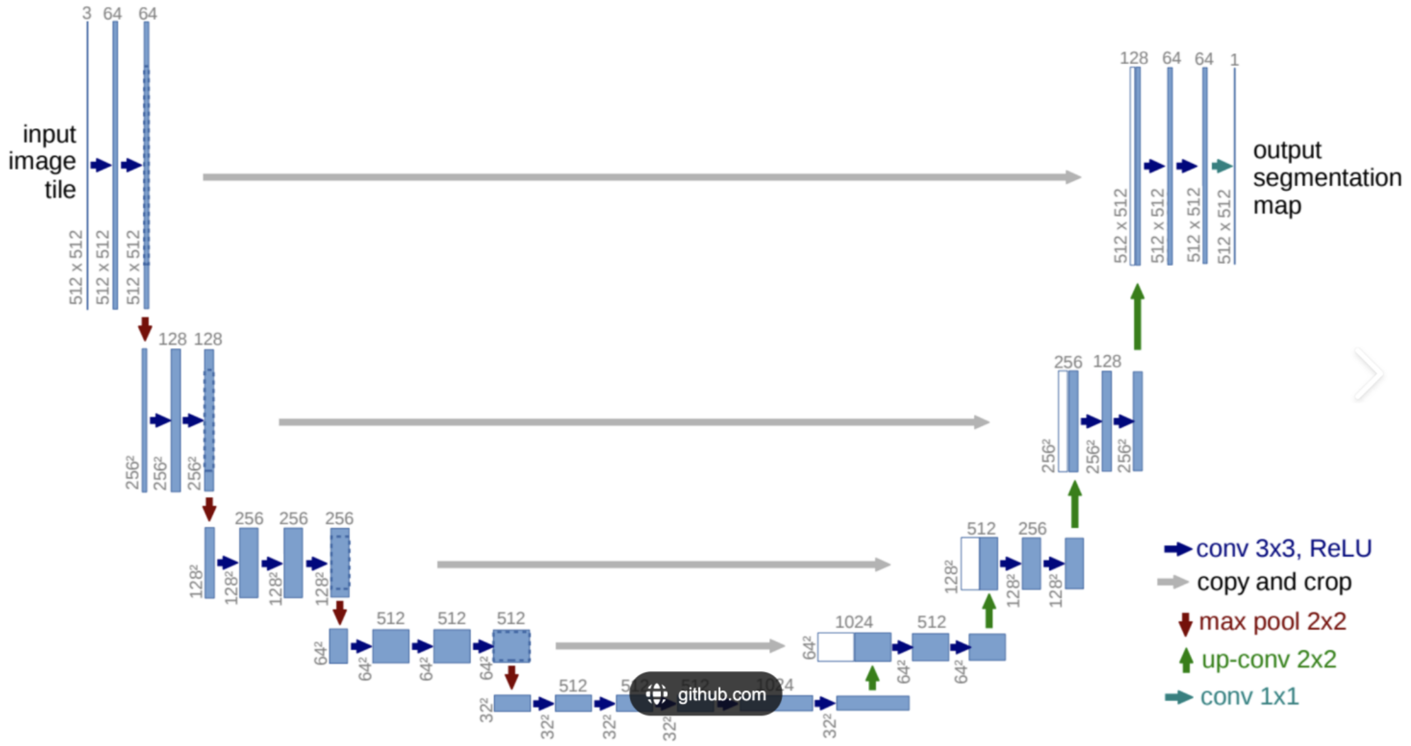


Semantic SLAM

Semantic Segmentation

U-Net

- The first half is dedicated to feature extraction, while the second half involves upsampling.
- **Encoder:** The left part consists of a downsampling module, which is composed of two 3x3 convolutional layers (with ReLU activation) followed by a 2x2 max pooling layer.
- **Decoder:** The right part is repeatedly structured with an upsampling convolutional layer (deconvolutional layer), feature concatenation (concat), and two 3x3 convolutional layers (with ReLU activation).



Semantic SLAM

Semantic Segmentation

DeepLab v3

•**Atrous (Dilated) Convolution:**

- To explicitly control the resolution of feature maps and enlarge the receptive field without increasing the number of parameters.

•**Encoder-Decoder Structure:**

- **Encoder:** A backbone network (e.g., ResNet) with atrous convolution to extract rich semantic features.
- **Decoder:** A simple yet effective module to refine object boundaries and recover spatial details from the encoder's output.

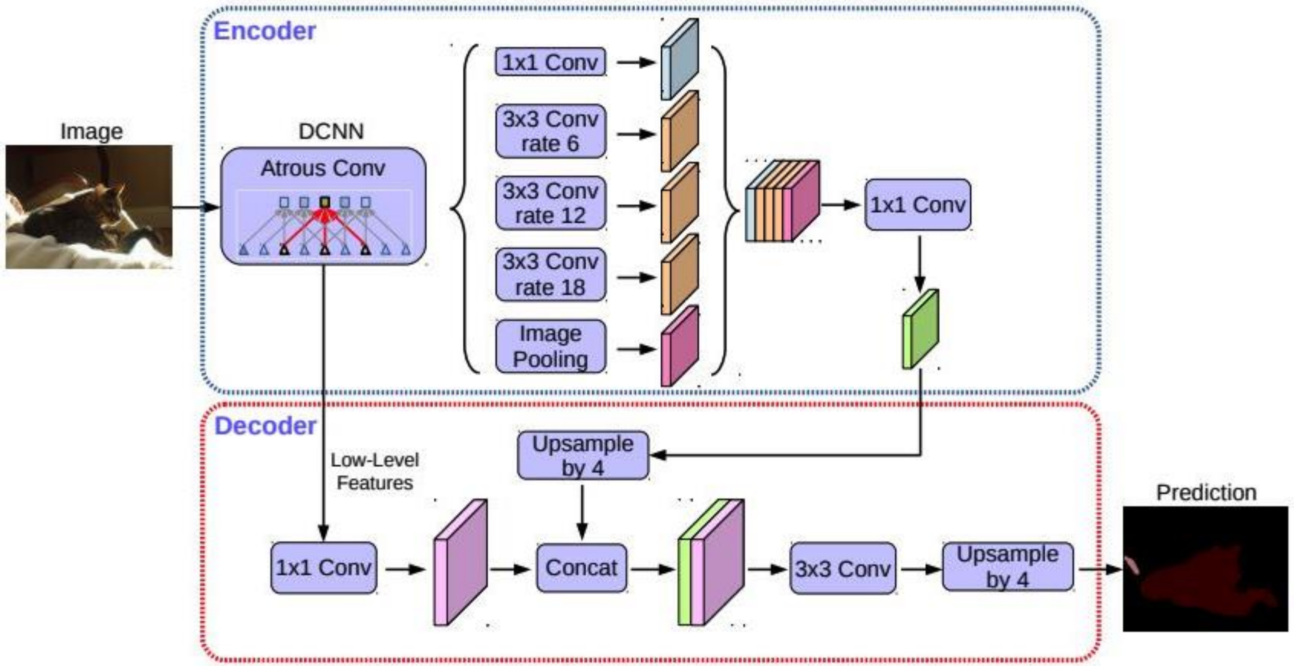


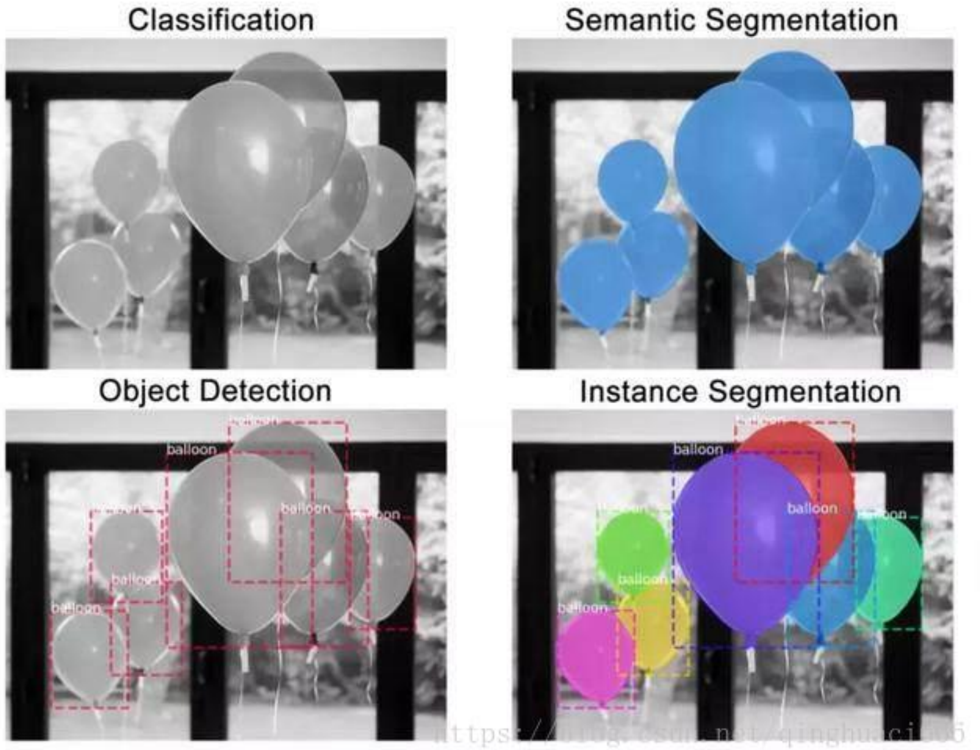
Fig. 2. Our proposed DeepLabv3+ extends DeepLabv3 by employing a encoder-decoder structure. The encoder module encodes multi-scale contextual information by applying atrous convolution at multiple scales, while the simple yet effective decoder module refines the segmentation results along object boundaries.

Semantic SLAM

Instance Segmentation:

1.Object Detection: Locating all objects in an image with a bounding box and assigning a class label.
2.Semantic Segmentation: Classifying every pixel in an image into a general category.

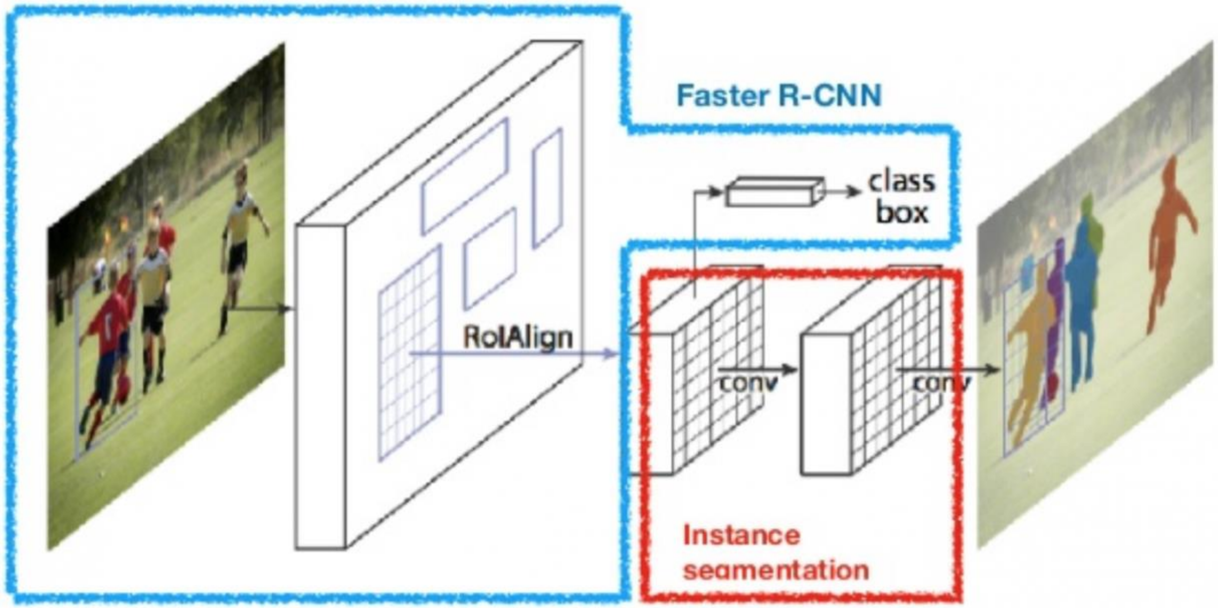
✓Mask-RCNN
✓YoLO



Semantic SLAM

Instance Segmentation:

✓Mask-RCNN



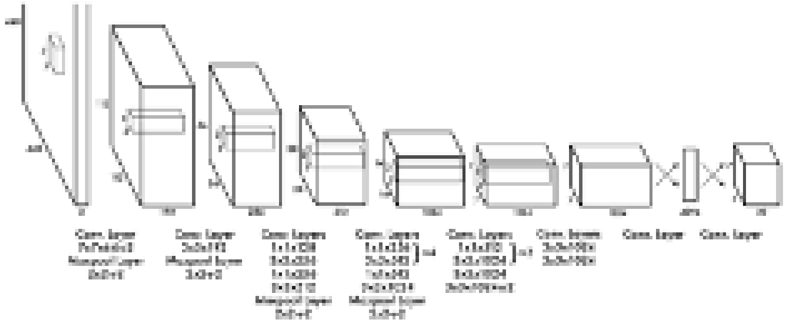
Two-stage

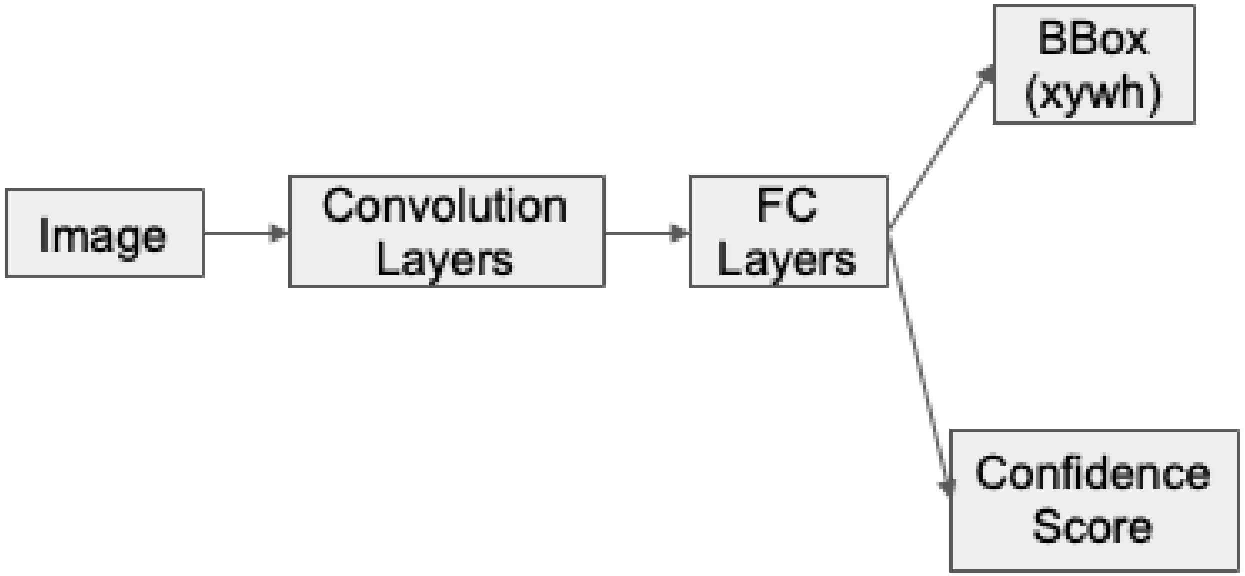
- Region proposal - get regions-of-interest (RoIs) - Region Proposal Network (RPN)
  - Run once per image
- Do classification and regression on the RoIs
  - Run on every RoIs

Semantic SLAM

Instance Segmentation:

- **YOLO**
  - Key concept: Frame object detection as a single regression problem from image pixels to bounding box coordinates and class probabilities.
  - an end-to-end (image→2d-bbox) regression problem ("unified detection")
  - one-stage detector

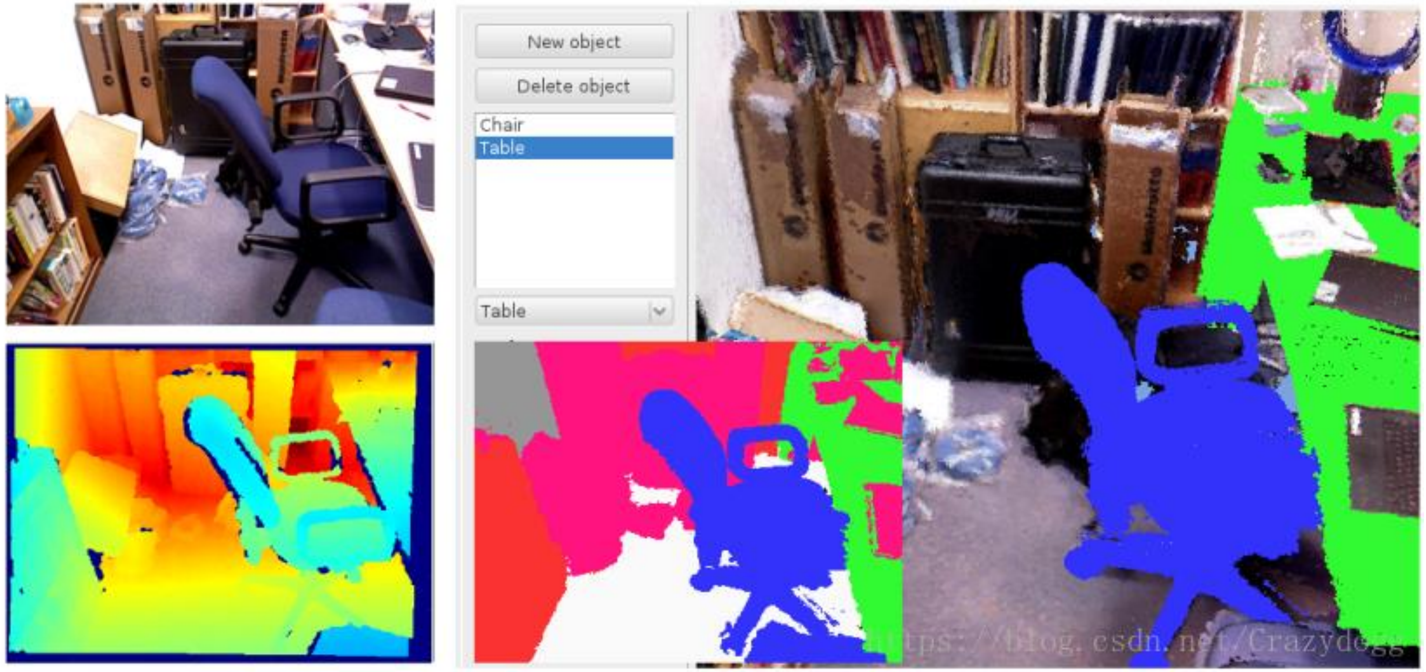




Semantic SLAM

SemanticFusion: Dense 3D Semantic Mapping with Convolutional Neural Networks (2017)

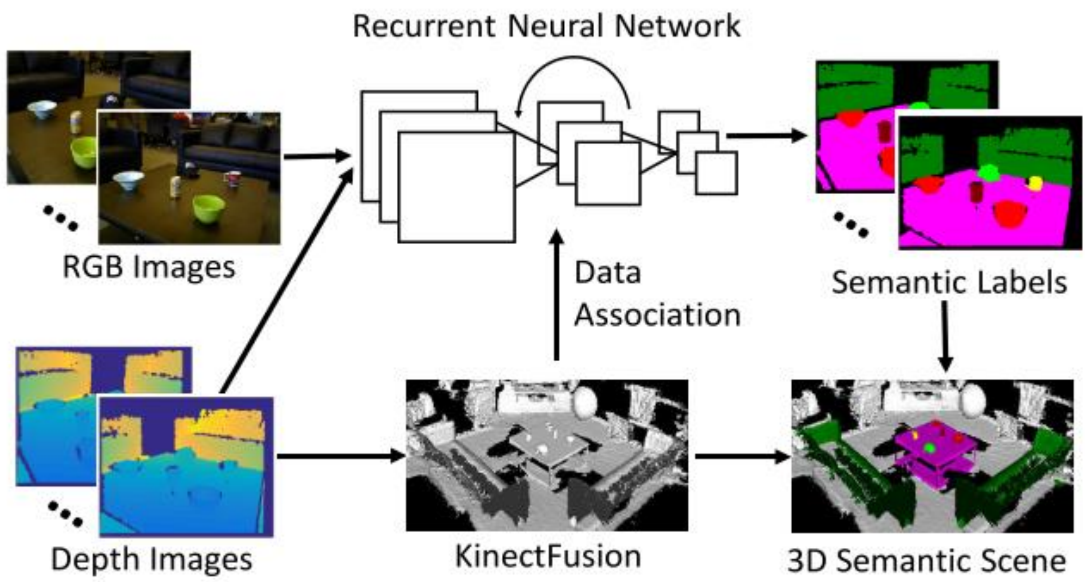
- The CNN takes 2D images as input and returns a probability distribution of classifications for each pixel.
- A Bayesian update model is used to track the classification probability distribution of each surface.
- The data association provided by the SLAM system is utilized to update these probabilities based on the CNN's predictions.
- Finally, a CRF framework is applied to leverage the scale information of the map itself to refine the semantic predictions.

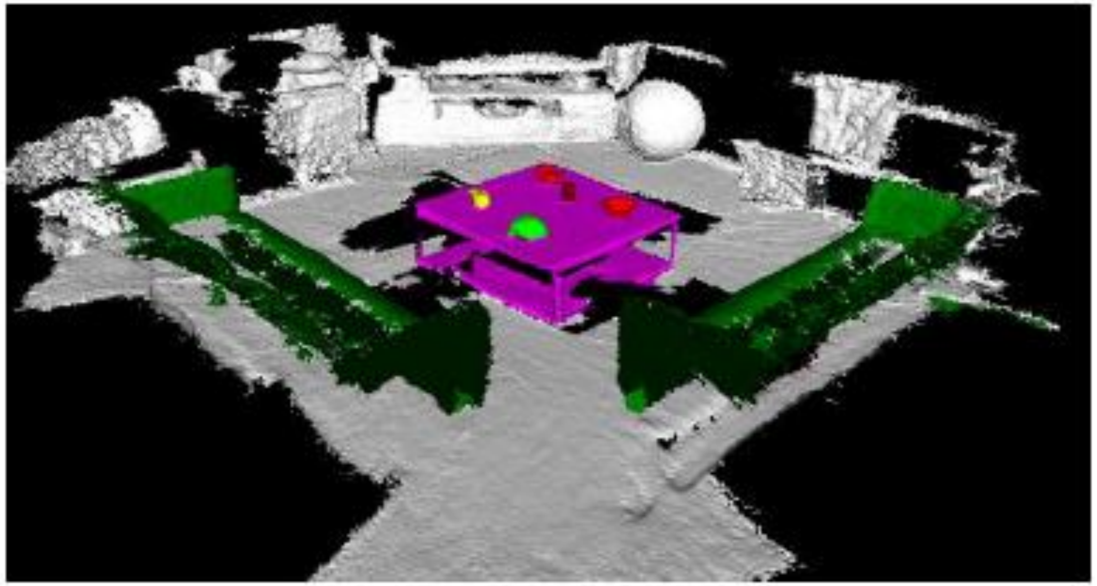


Semantic SLAM

DA-RNN: Semantic mapping with Data Associated Recurrent Neural Networks (2017)

- Uses RNN to associate and optimize semantic labels.
- KinectFusion provides the graph and data association.

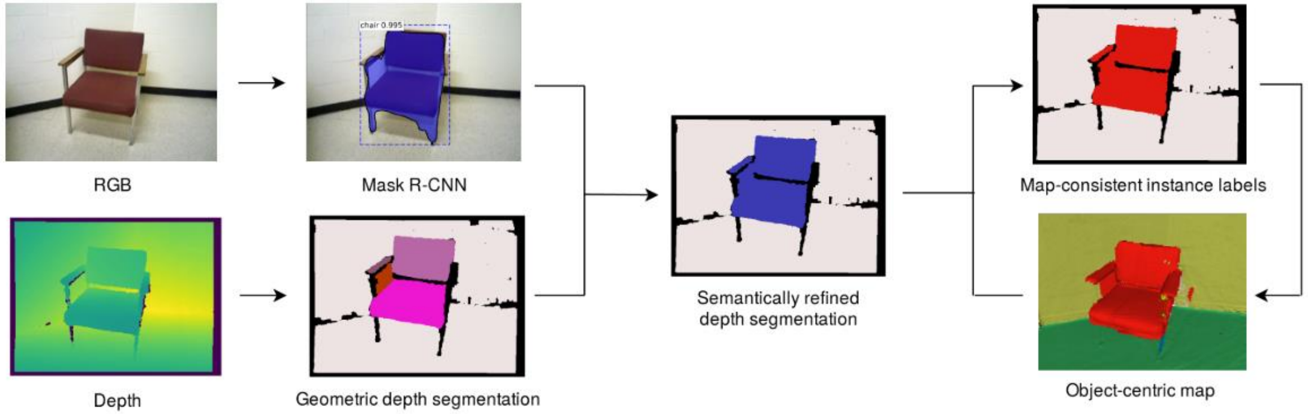




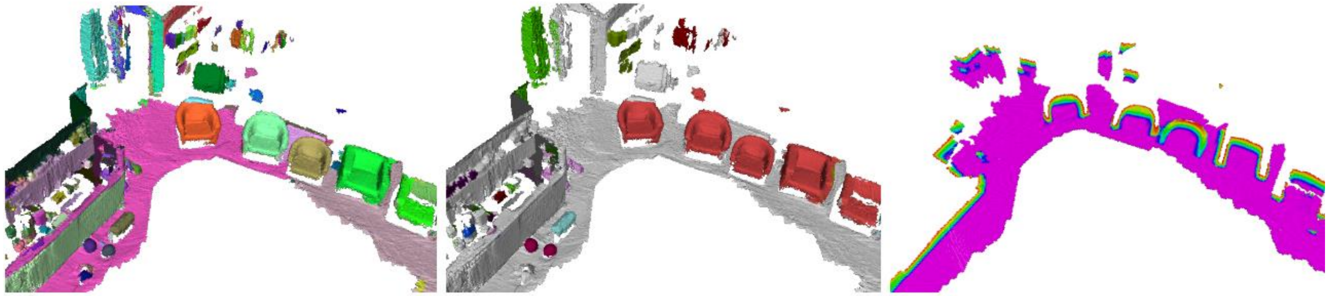
Semantic SLAM

Volumetric Instance-Aware Semantic Mapping and 3D Object Discovery (2019)

- A frame-to-frame segmentation framework combined with an unsupervised geometry-based method featuring instance-aware semantic prediction is used to simultaneously detect both recognized scene elements and previously unseen objects.
- The data-association step tracks predicted object instances across different frames.
- A map integration strategy fuses information about their 3D shape, position, and semantic data into a global voxel map.



(a) (b)

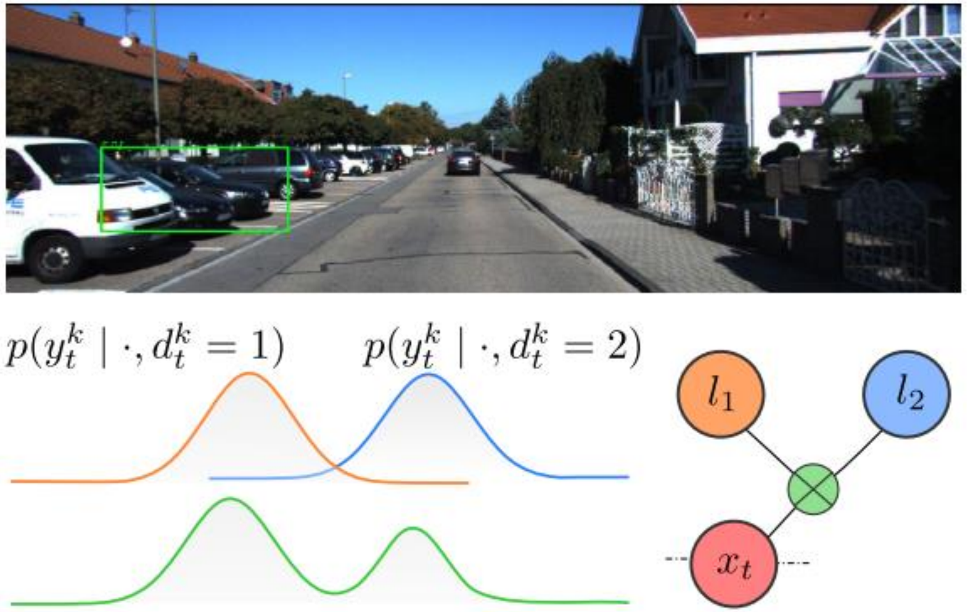


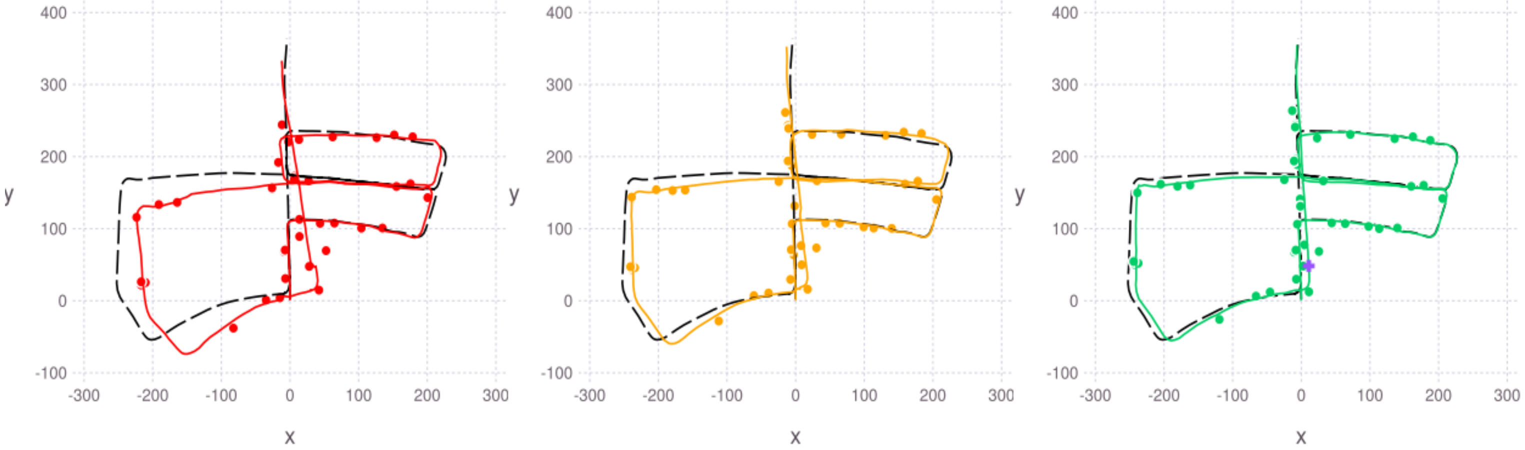
(c) (d) (e)

Semantic SLAM

Multimodal Semantic SLAM with Probabilistic Data Association (2019)

- Aiming at the semantic SLAM problem with ambiguous data association, this work proposes a nonparametric belief propagation method for full posterior inference.
- It describes multimodal semantic factors, incorporating uncertainties in data association and semantics as non-Gaussian factors into the factor graph, and uses mm-iSAM to solve the continuous optimization of poses and landmarks.



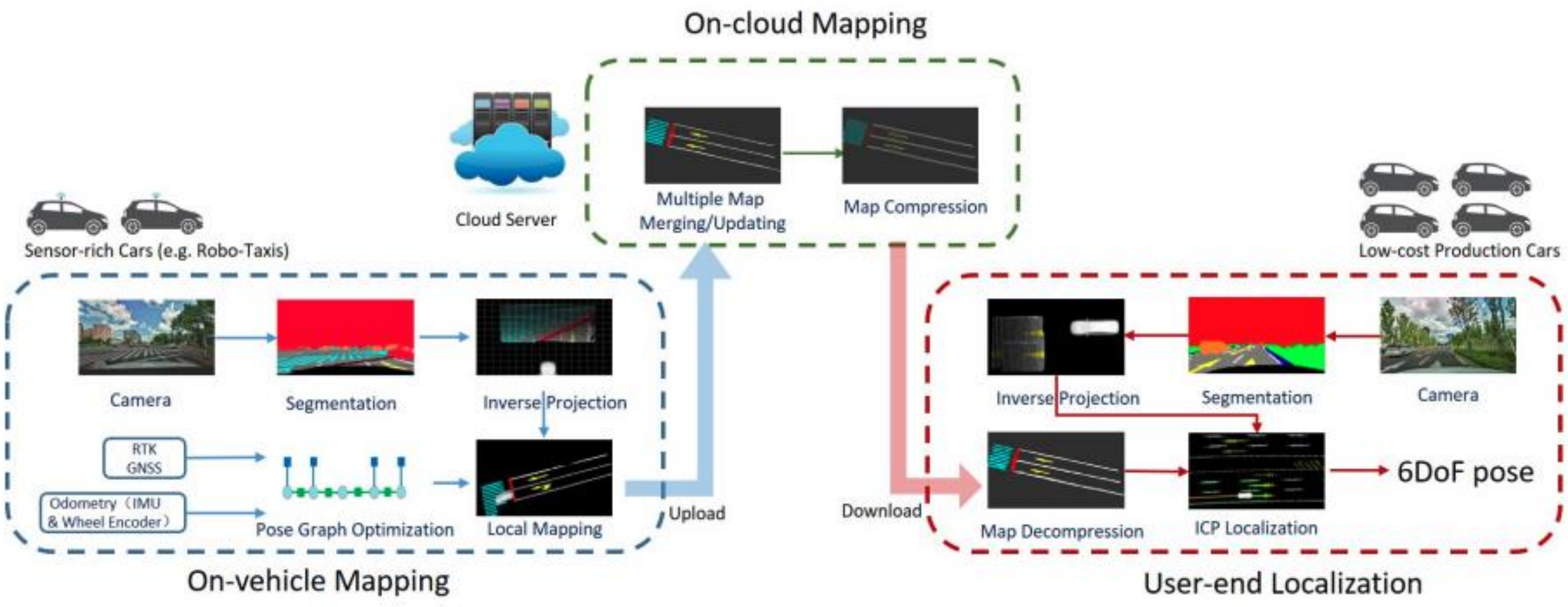


(a) Maximum-Likelihood (b) Gaussian PDA (c) MMSS (Ours)

Semantic SLAM

RoadMap: A Light-Weight Semantic Map for Visual Localization towards Autonomous Driving (2021)

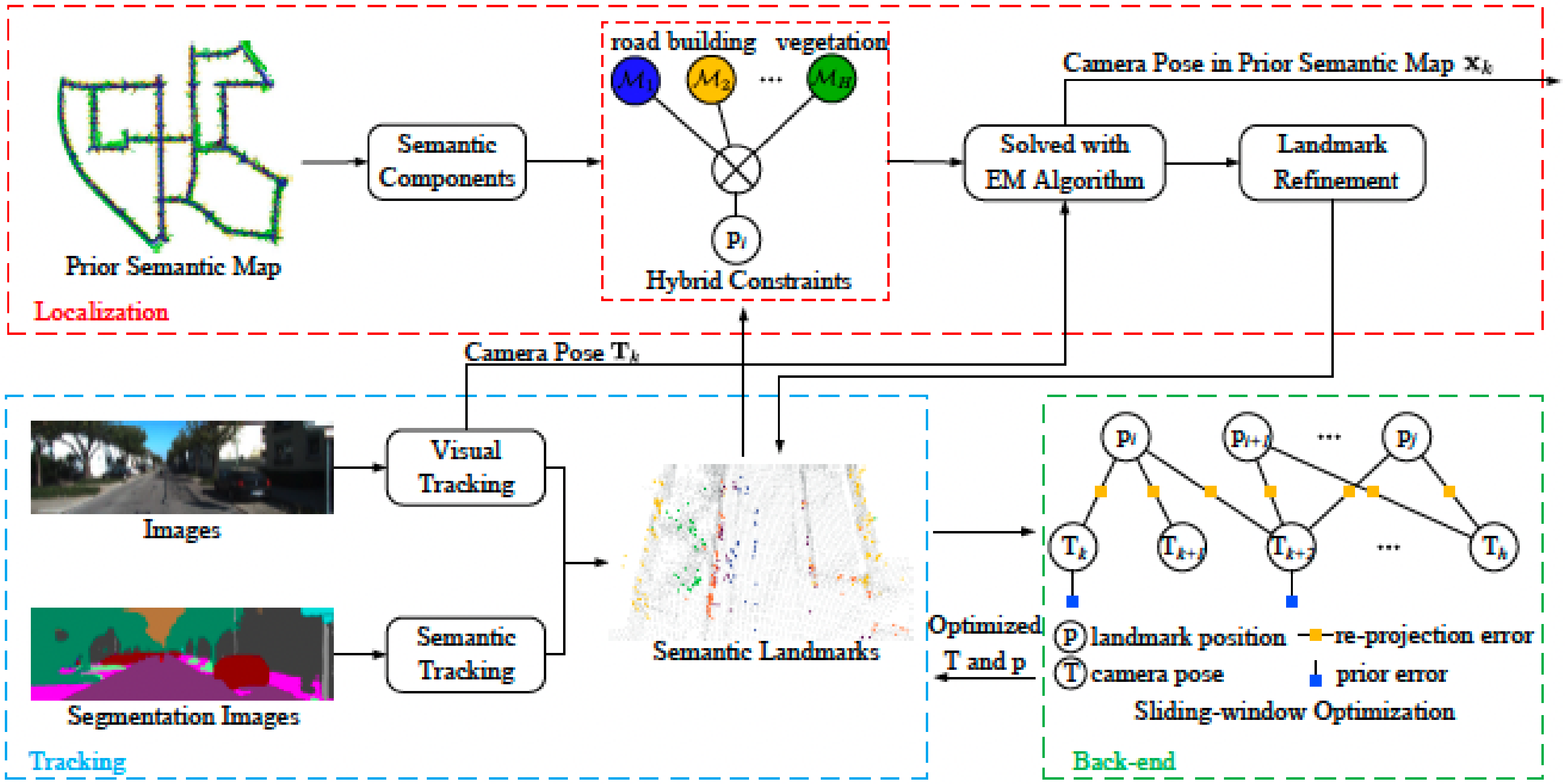
- Proposes a lightweight autonomous driving localization architecture: mapping on the vehicle side, maintenance in the cloud, and localization on the client side.
- For localization, semantic features are extracted from the front-facing camera and compared with the downloaded semantic map to determine the current vehicle's pose.



Semantic SLAM

SemLoc: Accurate and Robust Visual Localization with Semantic and Structural Constraints from Prior Maps (2022)

Utilizes semantic maps as constraints to achieve pose optimization for both relocalization and visual odometry.



Summary of Semantic SLAM

Semantic SLAM enhances SLAM by integrating object-level semantic information into the mapping and localization process. Instead of representing the world with just geometric primitives (points, lines, planes), it creates a map populated with recognizable, meaningful objects (e.g., chairs, cars, doors).

- **Robustness and Accuracy**: Use the stable, recognizable nature of objects to reduce drift in localization and enable more meaningful data association across different viewpoints.
- **High-Level Understanding**: Move from a "dense but dumb" 3D point cloud to a sparse but semantically rich map that robots can reason about (e.g., "navigate to the desk").
- **Human-Robot Interaction**: Enable robots to understand and execute commands based on object semantics (e.g., "go to the refrigerator").

Paradigms and Evolution

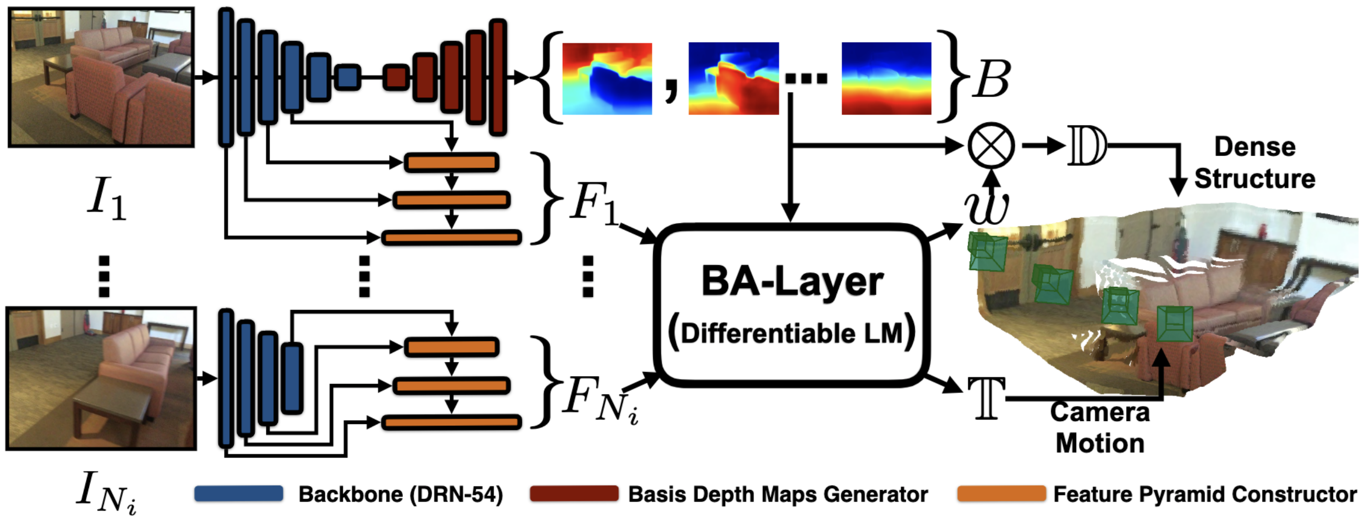
- **Early/Object-Based:** Represents the world as a collection of semantic objects (cubes, ellipsoids). Sparse but efficient.
- **Dense Semantic SLAM:** Creates dense 3D maps where every point/voxel has a semantic label. Computationally heavy but very detailed.
- **Probabilistic Frameworks:** Modern approaches (like the one in your previous query) use factor graphs to model the uncertainty in both geometry and semantics, leading to more robust systems.

Back-End Optimization

BA-Net: Dense bundle adjustment network

BA-Net integrates a differentiable second-order optimizer (LM algorithm) into a DNN for end-to-end learning.

- Instead of minimizing geometric or photometric error, BA-Net is performed on feature space to optimize the consistency loss of features from multiview images extracted by ConvNets.
- This feature-level optimizer can mitigate the fundamental problems of geometric or photometric solutions (e.g., some information may be lost in the geometric optimization, while environmental dynamics and lighting changes may impact the photometric optimization).
- This work combines domain knowledge of SLAM with deep learning and achieves successful results on large-scale real data, outperforming conventional SLAM with geometric or photometric BA and deep-learning-based methods



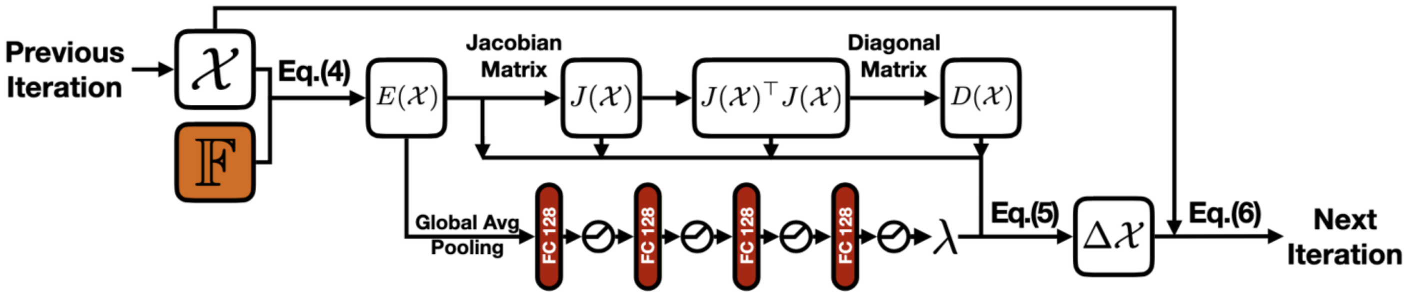


Figure 4: A single iteration of the differentiable LM.

Back-End Optimization

BA-Net: Dense bundle adjustment network

Advantages Over Conventional Methods

- **Solves Geometric BA Limitations:**
  - Less prone to losing high-level information in complex scenes.
- **Solves Photometric BA Limitations:**
  - Robust against lighting changes and dynamic objects.
- **Combines Strengths:** Merges SLAM domain knowledge with deep learning's pattern recognition.

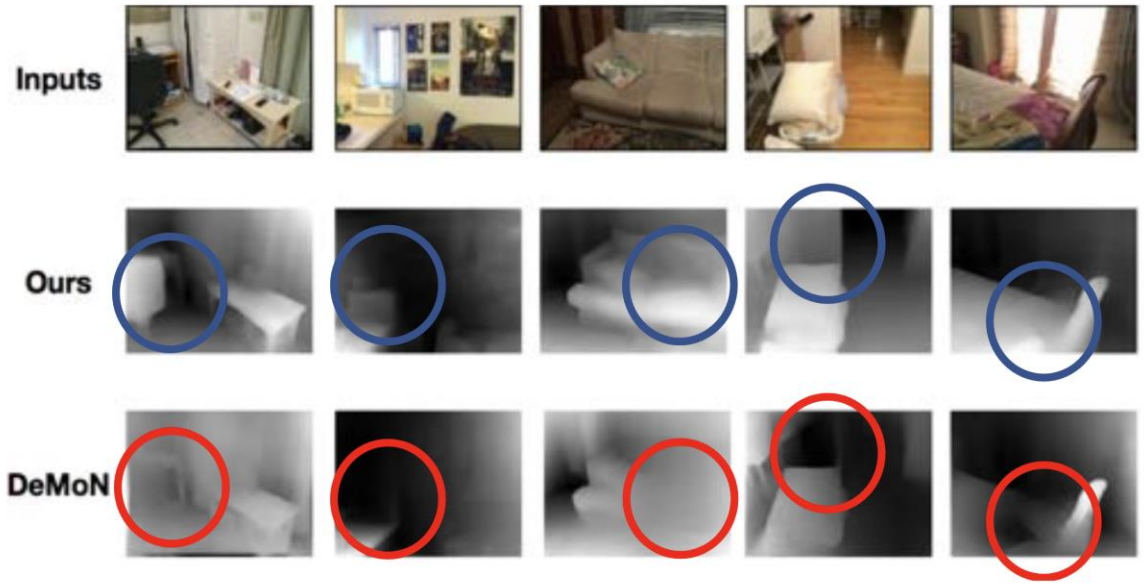


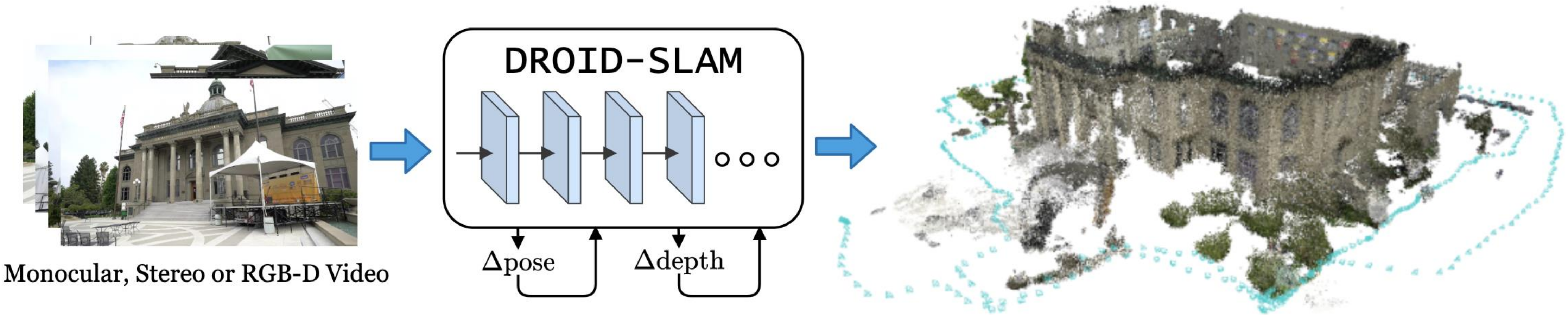
Figure 9: Qualitative Comparisons with DeMoN (Ummenhofer et al., 2017) on ScanNet.

	Ours	Ours*	DeMoN*	Photometric BA	Geometric BA
Rotation (degree)	**1.018**	1.587	3.791	4.409	8.56
Translation (cm)	**3.39**	10.81	15.5	21.40	36.995
Translation (degree)	**20.577**	31.005	31.626	34.36	39.392
abs relative difference	**0.161**	0.238	0.231	0.268	0.382
sqr relative difference	**0.092**	0.176	0.520	0.427	1.163
RMSE (linear)	**0.346**	0.488	0.761	0.788	0.876
RMSE (log)	**0.214**	0.279	0.289	0.330	0.366
RMSE (log, scale inv.)	**0.184**	0.276	0.284	0.323	0.357

Back-End Optimization

DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras (2021)

•Fully differentiable SLAM system unrolling iterative optimization.
•Maintains a **dense spatiotemporal graph** of keyframes.
•Uses learned **correlation volumes** to create data-driven geometric residuals.
•Iterative **pose + depth refinement** using differentiable Gauss-Newton updates.



Back-End Optimization

DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras (2021)

Dense Bundle Adjustment Layer (DBA) The Dense Bundle Adjustment Layer (DBA) maps the set of flow revisions into a set of pose and pixelwise depth updates. We define the cost function over the entire frame graph

$$\mathbf{E}(\mathbf{G}', \mathbf{d}') = \sum_{(i,j)\in\mathcal{E}} \left\| \mathbf{p}_{ij}^* - \Pi_c(\mathbf{G}'_{ij} \circ \Pi_c^{-1}(\mathbf{p}_i, \mathbf{d}'_i)) \right\|^2_{\Sigma_{ij}} \qquad \Sigma_{ij} = \operatorname{diag} \mathbf{w}_{ij}. \tag{4}$$

where $\|\cdot\|_\Sigma$ is the Mahalanobis distance which weights the error terms based on the confidence weights $\mathbf{w}_{ij}$. Eqn. 4 states that we want an updated pose $\mathbf{G}'$ and depth $\mathbf{d}'$ such that reprojected points match the revised correspondence $\mathbf{p}_{ij}^*$ as predicted by the update operator.

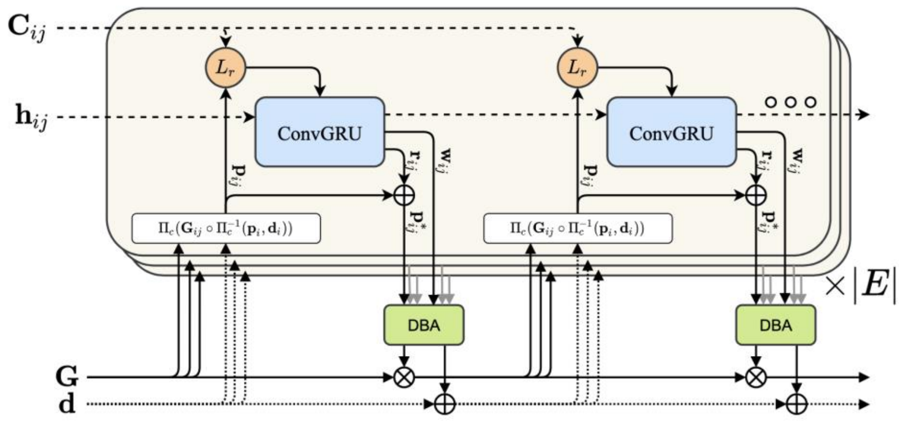
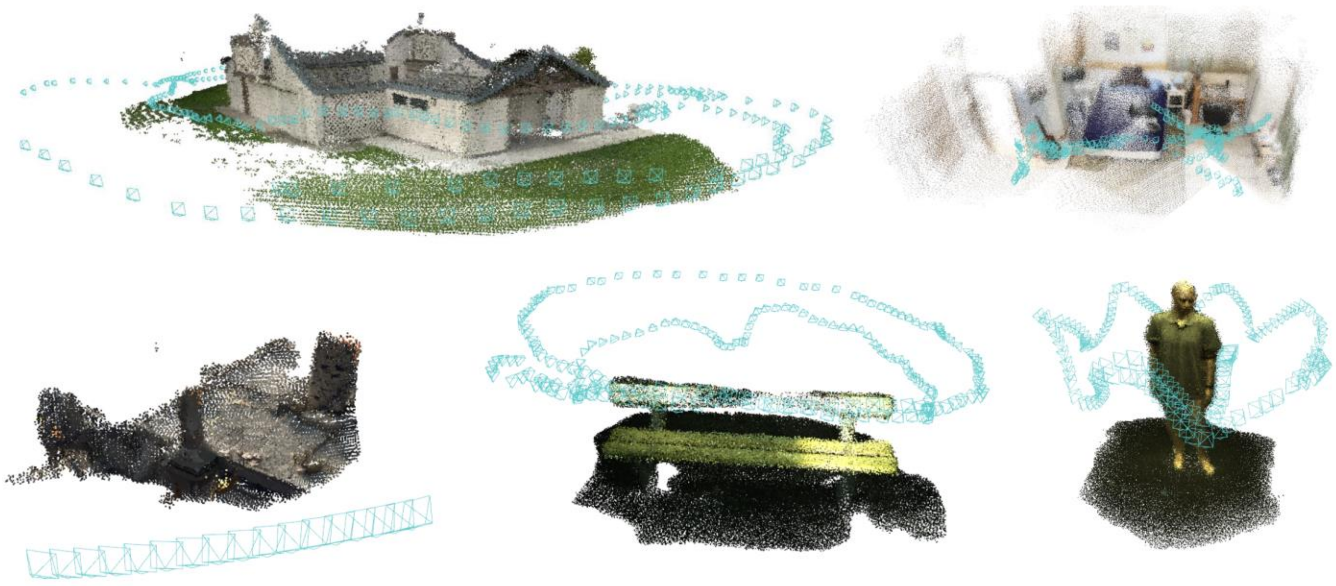


Figure 2: Illustration of the update operator. The operator acts on edges in the frame graph, predicting flow revisions which are mapped to depth and pose update through the (DBA) layer.

Back-End Optimization



		MH01	MH02	MH03	MH04	MH05	V101	V102	V103	V201	V202	V203	Avg
Deep/Hyb.	DeepFactors [9]	1.587	1.479	3.139	5.331	4.002	1.520	0.679	0.900	0.876	1.905	1.021	2.040
	DeepV2D [47]†	0.739	1.144	0.752	1.492	1.567	0.981	0.801	1.570	0.290	2.202	2.743	1.298
	DeepV2D (Tartan Air)†	1.614	1.492	1.635	1.775	1.013	0.717	0.695	1.483	0.839	1.052	0.591	1.173
	TartanVO[1] [53]†	0.639	0.325	0.550	1.153	1.021	0.447	0.389	0.622	0.433	0.749	1.152	0.680
	D3VO + DSO [57]†	-	-	0.08	-	0.09	-	-	0.11	-	0.05	0.19	-
Classical	ORB-SLAM [31]	0.071	0.067	0.071	0.082	0.060	**0.015**	0.020	X	0.021	0.018	X	-
	DSO [15]†	0.046	0.046	0.172	3.810	0.110	0.089	0.107	0.903	0.044	0.132	1.152	0.601
	SVO [18]†	0.100	0.120	0.410	0.430	0.300	0.070	0.210	X	0.110	0.110	1.080	-
	DSM [60]	0.039	0.036	0.055	0.057	0.067	0.095	0.059	0.076	0.056	0.057	0.784	0.126
	ORB-SLAM3 [5]	0.016	0.027	0.028	0.138	0.072	0.033	0.015	0.033	0.023	0.029	X	-
	Ours (odometry only)†	0.163	0.121	0.242	0.399	0.270	0.103	0.165	0.158	0.102	0.115	0.204	0.186
	Ours	**0.013**	**0.014**	**0.022**	**0.043**	**0.043**	0.037	**0.012**	**0.020**	**0.017**	**0.013**	**0.014**	**0.022**

- Differentiable back-end optimization bridges the gap between classic geometry and deep learning by making the SLAM optimizer itself trainable.
- Methods like **BA-Net** and **DROID-SLAM** demonstrate that learned optimization improves robustness and accuracy, but challenges remain in scalability, efficiency, and generalization.

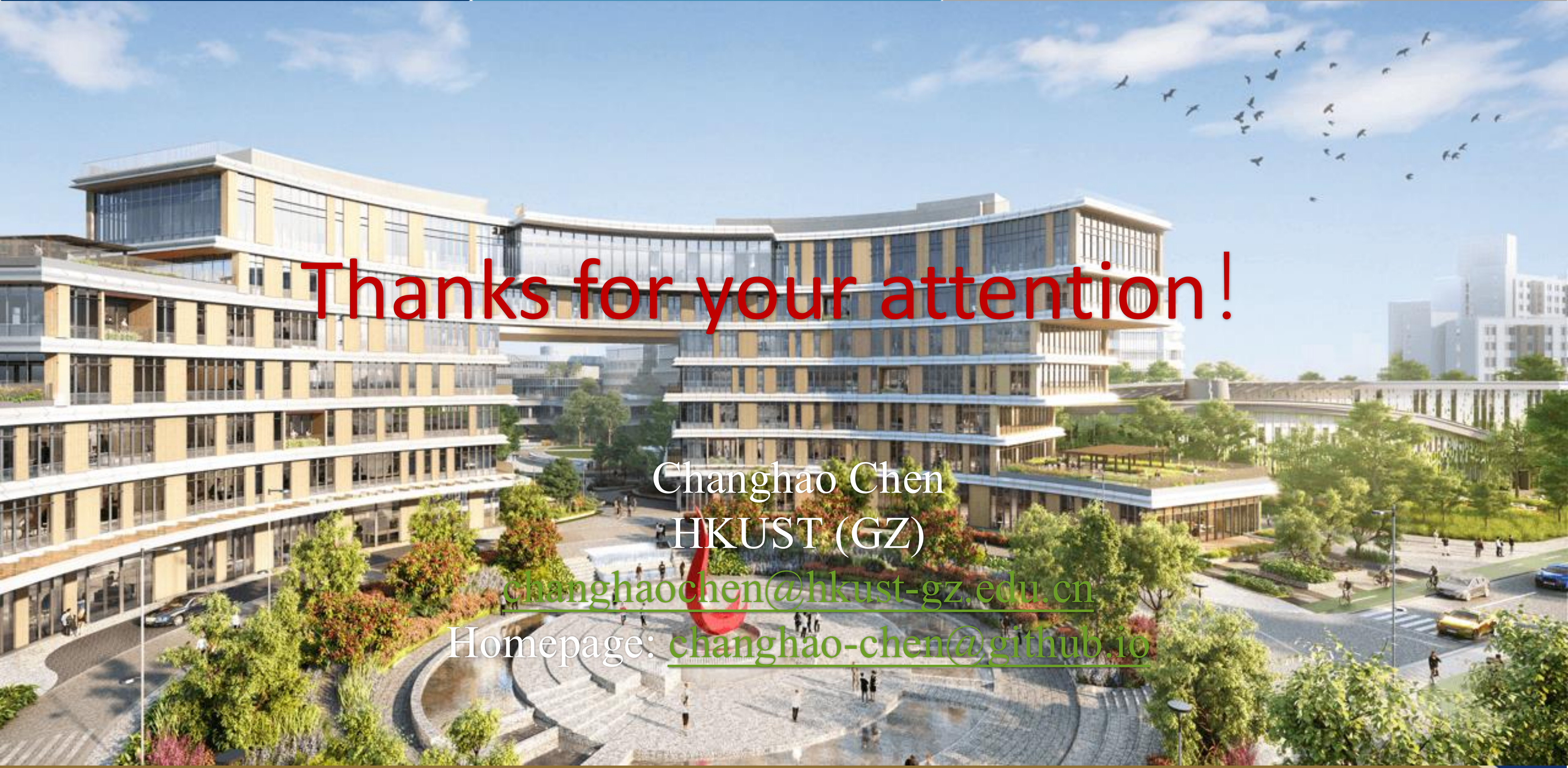
Summary

- Deep learning enhances SLAM in:
  - **Front-end: features, pose estimation**
  - **Back-end: differentiable optimization**
  - **Relocalization and semantic mapping**

- Challenges remain: efficiency, interpretability, long-term robustness

Future Directions

- Joint SLAM + perception models
- Continual learning for long-term navigation
- Lightweight, real-time deep SLAM on vehicles
- Integration with large-scale map priors and cloud computing



Thanks for your attention！

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